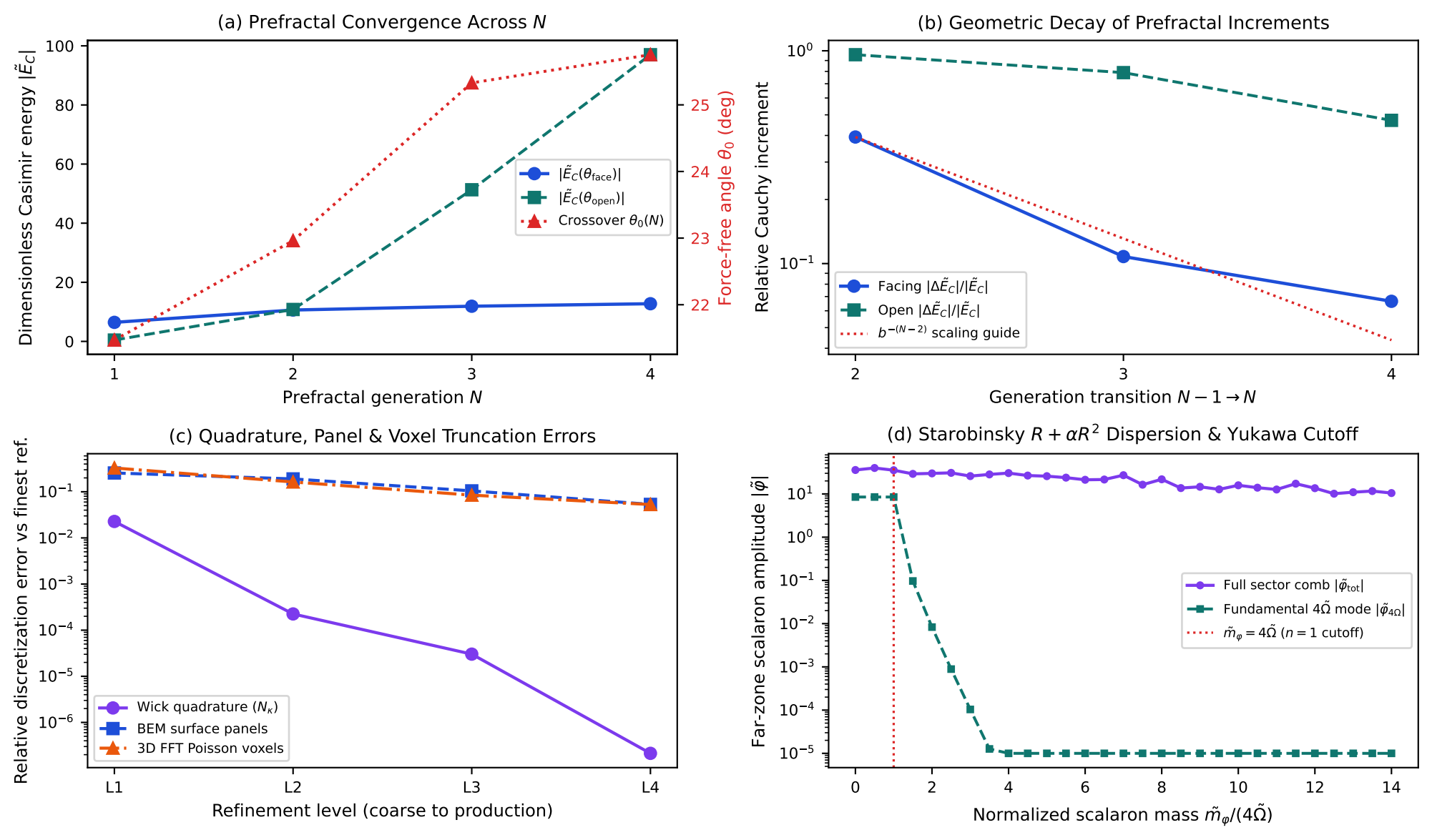
(a) Prefractal Convergence Across N
Dimensionless Casimir energy $|\tilde{E}_C|$
Force-free angle θ_0 (deg)
Prefractal generation N
$|\tilde{E}_C(\theta_{\text{face}})|$
$|\tilde{E}_C(\theta_{\text{open}})|$
Crossover $\theta_0(N)$
(b) Geometric Decay of Prefractal Increments
Relative Cauchy increment
Generation transition $N-1 \to N$
Facing $|\Delta\tilde{E}_C|/|\tilde{E}_C|$
Open $|\Delta\tilde{E}_C|/|\tilde{E}_C|$
$b^{-(N-2)}$ scaling guide
(c) Quadrature, Panel & Voxel Truncation Errors
Relative discretization error vs finest ref.
Refinement level (coarse to production)
Wick quadrature (N_κ)
BEM surface panels
3D FFT Poisson voxels
(d) Starobinsky $R + \alpha R^2$ Dispersion & Yukawa Cutoff
Far-zone scalaron amplitude $|\tilde{\varphi}|$
Normalized scalaron mass $\tilde{m}_\varphi/(4\tilde{\Omega})$
Full sector comb $|\tilde{\varphi}_{\text{tot}}|$
Fundamental $4\tilde{\Omega}$ mode $|\tilde{\varphi}_{4\Omega}|$
$\tilde{m}_\varphi = 4\tilde{\Omega}$ ($n=1$ cutoff)